

Rishi Chilveri

Worli, Mumbai

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EDUCATION

Bharati Vidyapeeth (Deemed to Be University)

Bachelor of Engineering in Computer Science — CGPA: 9.64

Navi Mumbai, Maharashtra

2022 – 2026

WORK EXPERIENCE

Software Developer Intern

Noice Innovation Labs Pvt. Ltd., Mumbai, India

June 2025 – August 2025

- Designed and Optimized a high-capacity data encoding system in C.
- Ported CLI-based barcode generation tools to a responsive web and mobile application, eliminating the need for local environment setup.
- Integrated REST APIs into internal tools, enabling generation of encrypted, high-capacity barcodes accessible only to authorized users with a decryption key.

PROJECTS

Self-Healing RAG - LangGraph, ChromaDB, Groq, BM25, Cross-Encoder, Streamlit (2026)

- Engineered a self-healing Retrieval-Augmented Generation pipeline using LangGraph, deploying a critic agent that validates answer groundedness and triggers automatic query reformulation, achieving a 0% hallucination rate on a self-built 30-question evaluation set.
- Implemented hybrid retrieval combining BM25 keyword search and vector similarity via Reciprocal Rank Fusion, followed by cross-encoder reranking, improving answer accuracy from 92% (baseline RAG) to 96%.
- Built an automated evaluation harness benchmarking retrieval accuracy, hallucination rate, and latency across two RAG architectures, and deployed the system with multi-model fallback for uninterrupted inference under per-model rate limits.

Pixel Studio - React, FastAPI, ComfyUI, Stable Diffusion (2026)

- Engineered a full-stack text-to-image generator using Stable Diffusion, streaming results to a React frontend via FastAPI WebSocket.
- Achieved average generation time of 10–12s on consumer hardware with real-time progress updates to the UI.
- Applied attention slicing to optimize model execution on low-VRAM GPUs, reducing OOM crashes by 30%.

Smart Attendance System - React, FastAPI, PostgreSQL, InsightFace, FAISS, MediaPipe (2025)

- Architected a real-time face recognition attendance system scanning and marking students every 2.5 seconds using ArcFace embeddings.
- Indexed face embeddings with FAISS for sub-second (<50ms) vector search, enabling instant identity lookup across all registered students.
- Deployed liveness detection via MediaPipe to block photo spoofing, and enforced duplicate-prevention logic ensuring one entry per student per day.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript

Frontend: React.js, HTML, CSS

Backend & Databases: FastAPI, Flask, Django, PostgreSQL, MongoDB, MySQL

ML & AI: PyTorch, TensorFlow, CNN, ANN, Stable Diffusion, InsightFace, FAISS, MediaPipe, Transfer Learning, Computer Vision

Tools & DevOps: Git, Docker, WebSocket, ComfyUI

ACHIEVEMENTS & CERTIFICATIONS

- Convolutional Neural Networks** — DeepLearning.AI, authorized by Andrew Ng via Coursera (Jul 2025). [Verify](#)
- Mastering Generative AI and ChatGPT** — 5-week course by GeeksforGeeks. [Verify](#)
- Passed the **Python 3.4.3 Test** conducted by IIT Bombay's Spoken Tutorial Project with a score of 70%, earning 4 credits. [Verify](#)